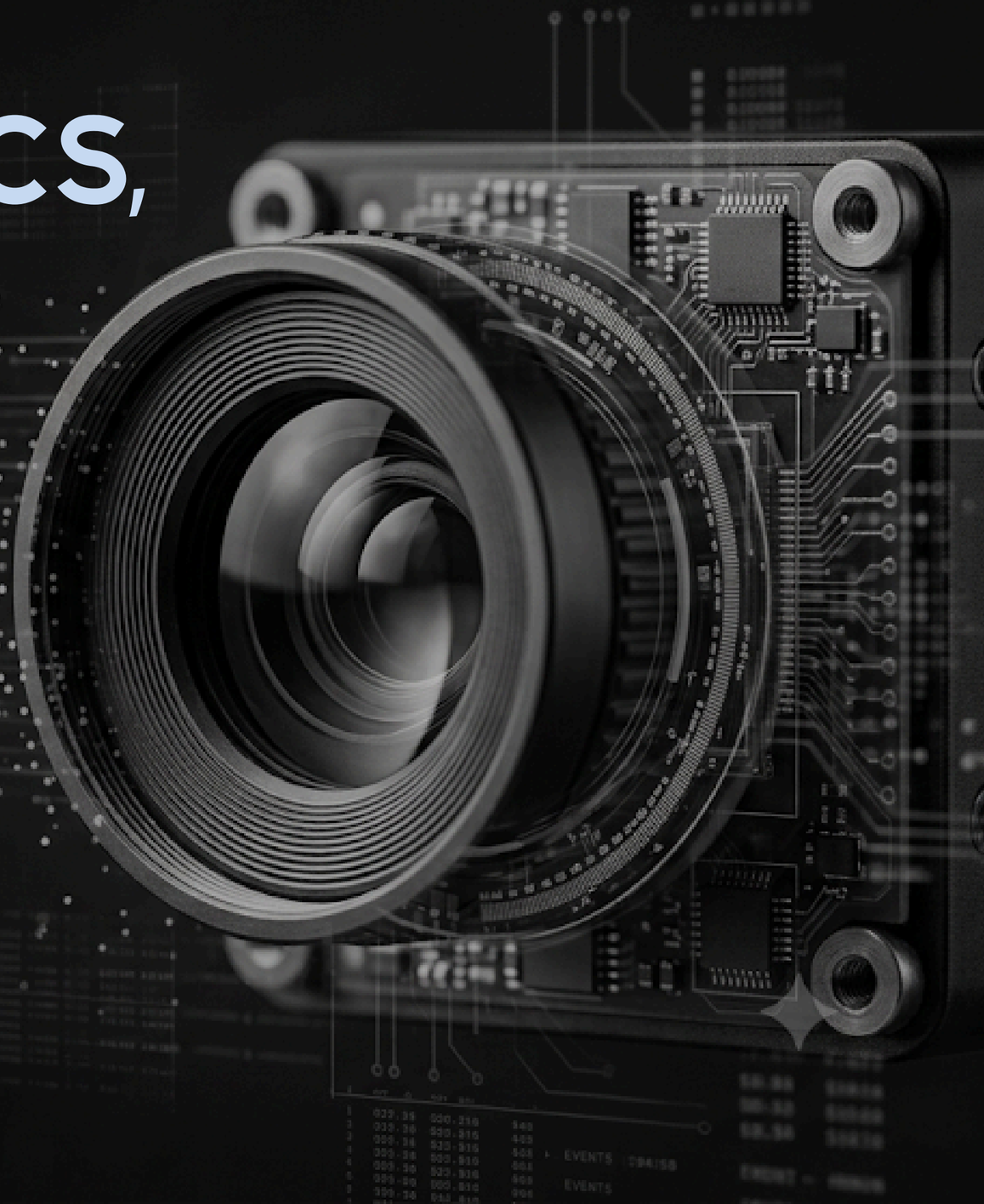
BIO-INSPIRED OPTICS, IMAGING & PHOTONICS

Event-Based Cameras Inspired by the Human Eye



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What are event based cameras?

An event-based camera (also Dynamic Vision Sensor) (DVS) or silicon retina, which is a neuromorphic sensor whose pixels independently and asynchronously respond to relative changes in brightness.

Free Frame Operation:

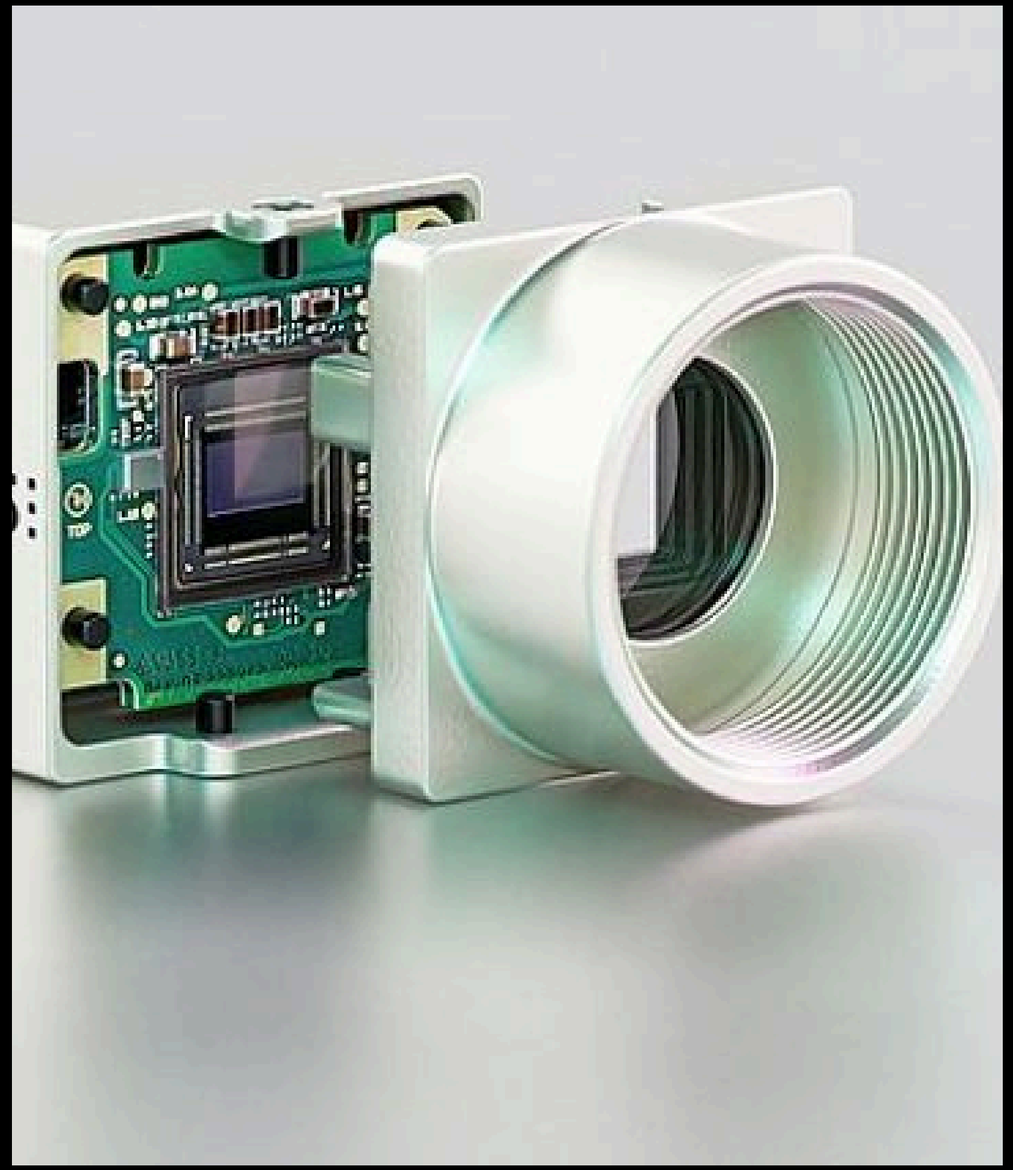
Instead of waiting for a global shutter clock to output dense two-dimensional matrices, each pixel fires immediately whenever local illumination changes by a set percentage.

Continuous Event Streams:

Output consists of a continuous, microsecond-resolution stream of digital address-events representing moving edges and intensity fluctuations.

Activity-Driven Sensing:

When a scene is completely static, the sensor stays silent; the faster an object moves, the more events are generated per second.



Why does this tech matter?

Since event-based cameras respond to changes in brightness of each pixel asynchronously, they can provide very low latency, ultra high dynamic range(>120db) and minimal motion blur.



The Bottleneck:

Conventional cameras impose a global clock across all pixels, forcing a trade-off among latency, data bandwidth, and dynamic range.

Motion Blur & Blind Spots:

Fast-moving targets smear across the sensor during fixed exposure windows and go completely unmonitored during the blind intervals between consecutive frames.

Frame-based Cameras:

Create lot of noise, motion blur, have a higher latency and have blind spots

Event-Based Cameras:

Maintain resolution while capturing fast-moving subjects clearly without blur even in dim light.

How Conventional **Frame-Based Cameras** Work

Consumer cameras (30–60 Hz) and specialized high-speed industrial cameras (up to 5,000 Hz) capture discrete static snapshots at uniform time intervals (16.7ms apart at 60fps).



- **Single Exposure Limitation:** Because every pixel in a standard frame shares the same exposure time, scenes containing both deep shadows and bright highlights cause underexposure in dark areas and photodiode saturation in bright regions.
- **Limited Dynamic Range:** Standard frame-based sensors are restricted to roughly 60 dB of dynamic range (a 1,000:1 linear light ratio).
- **Aliasing Under Rapid Motion:** A 60 fps camera has a 30 Hz Nyquist limit, causing severe temporal aliasing and motion blur when observing high-frequency vibrations or fast projectiles.



The Human Eye - The Biological Foundation

The human eye detects changes in light intensity because it functions as an asynchronous, continuous stream of electrochemical signals processed by rod and cone photoreceptors. This allows for high dynamic range through logarithmic perception, enabling the brain to map luminance ranges exceeding 5,000:1.

- **Continuous Signal Processing:** Vision is a constant flow of data where rods (low light/peripheral) and cones (colour/detail) reset asynchronously, avoiding global shutter delays.
- **Logarithmic Response:** The brain perceives brightness logarithmically, making it highly sensitive to relative changes in intensity rather than absolute linear values
- **Peripheral Sensitivity**: The periphery, dominated by rods, has a higher flicker fusion threshold and faster response time than the foveal cone-dominated center, enhancing rapid motion detection.



A close up of the human eye

How the Retina Processes Visual Information

The retina converts light to nerve signals, it is the reason we can see in different conditions, in such a dynamic way. Its connection to the brain is mediated via the optic nerve.

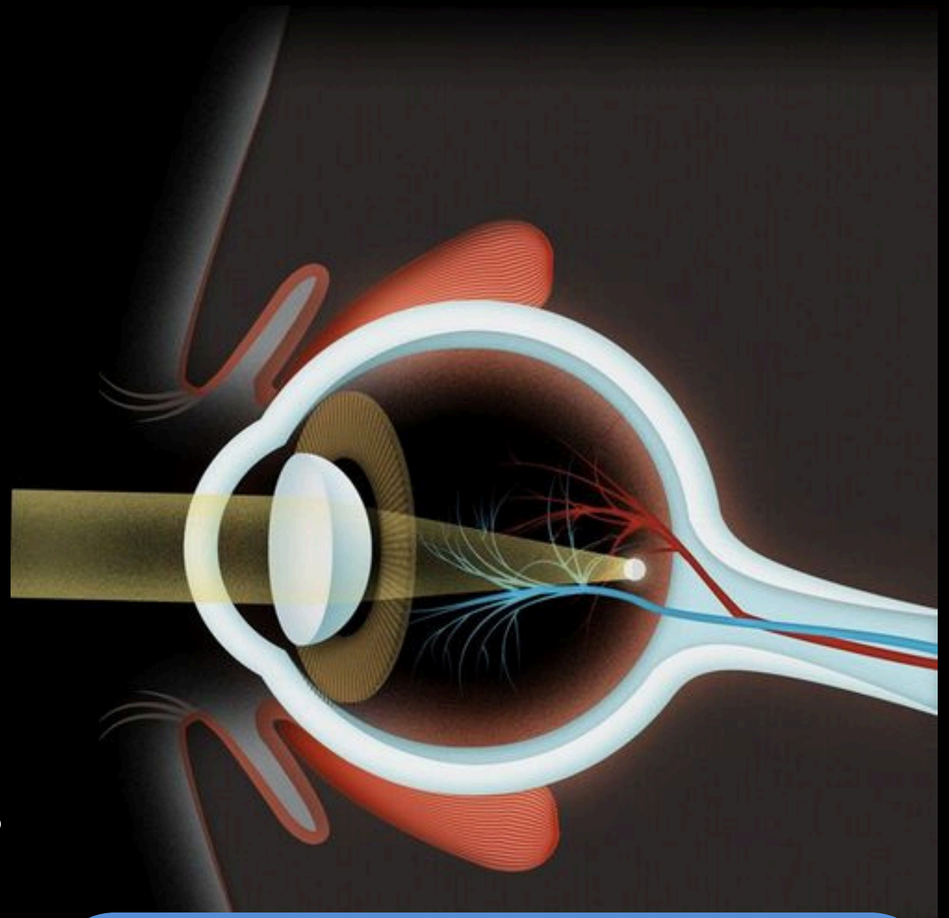
The retina is made of four main parts:

- **Photoreceptor Cells:** Rods - Responsible for vision in low light
  Cones - Responsible for vision in high light
  Bipolar Cells - Responsible for transmitting signals from rods and cones to ganglion cells

- **Ganglion Cells:** The final layer of cells in the retina, ganglion cells send the visual information to the brain via the optic nerve.

- **Macula:** This is the central part of the retina, responsible for our sharpest and most detailed vision. It has a high quantity of cones, making it responsible for activities such as reading and recognizing faces

- **Fovea:** The central part of the macula, with the highest density of cones. It helps us see the little and fine details



An inside view to the retina

Event-based Vision Sensor (EVS) - Detecting Changes in Moving Subjects [Technol

Sony



Event-Based Vision Sensor

Sony's EVS

- Technology -

Watch on YouTube

Biomimicry -Biological Retina vs Silicon DVS

Biomimicry is the practice of making technological and industrial design copy naturally refined processes.

Here, we are abstracting the human retina's core computational strengths into silicon circuitry.

- **Logarithmic Compression:** Both biological photoreceptors and DVS pixels apply front-end logarithmic conversion, enabling >120 dB to 140 dB dynamic range without saturation.
- **ON/OFF Polarity Channels:** Dual pixel comparators (DIFF_ON, DIFF_OFF) directly emulate transient ON and OFF retinal ganglion cells emitting sparse spike trains.
- **Pixel Independence:** Every DVS pixel is strictly independent in the analog array, lacking the lateral centre-surround coupling of biological horizontal and amacrine cells.

How An **Event Based Camera** Works?

1. The process behind event generation

- **Event Generation:**

Whenever a change occurs, the pixel outputs an "event", either **ON** or **OFF**, which contains four critical data points:

-**X, Y:** Horizontal and vertical pixel coordinates.

-**T:** Precise timestamp in microseconds.

-**P (Polarity):** Whether the light intensity increased (ON) or decreased (OFF).

Unchanged areas generate no events, reducing redundant visual data.

2. Mathematics behind event generation

$$L(x,y,t) = \ln I(x,y,t)$$

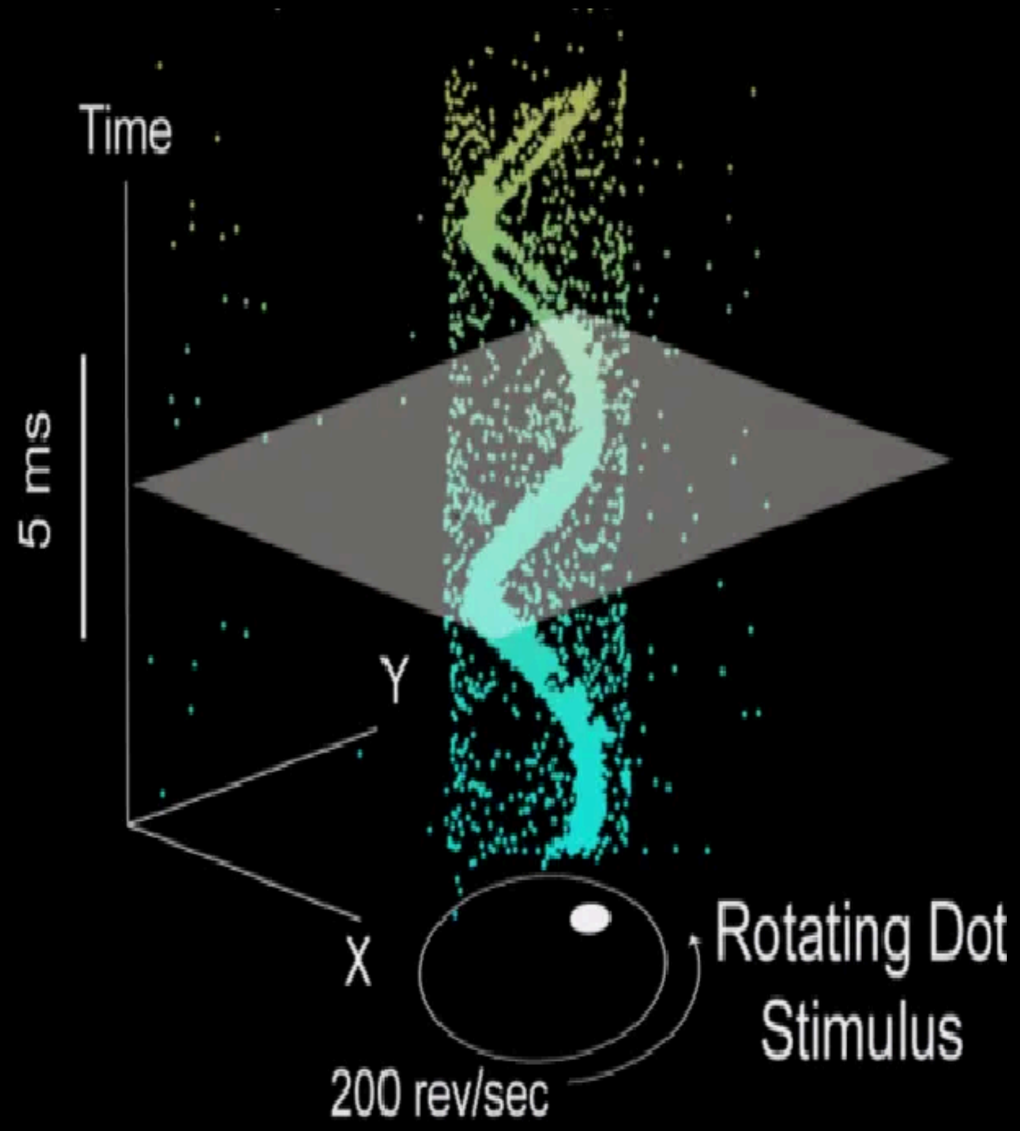
Logarithmic brightness:

$$L(x, y, t_k) - L(x, y, t_{last}) = pC$$

An event is generated when the change reaches the contrast threshold.

$$e_k = (x_k, y_k, t_k, p_k)$$

Where: I = light intensity, L = log intensity, C = contrast threshold, p = +1 for brighter (ON) and p = -1 for darker (OFF), and t_{last} = time of the previous event at that pixel.



An event stream for a rotating dot on a disc

Understanding the Event Stream

Spatiotemporal Representation of Scene Dynamics

- **Asynchronous Time-Domain Output:** Streams consist of discrete, asynchronous event packets rather than continuous, synchronous frames.
- **Pixel-Level Independence:** Every pixel operates independently, firing data packets only when change occurs.
- **Microsecond Temporal Resolution:** Captures precise timing data equivalent to >10,000 fps performance

Processing and Interpreting Event Streams

From Sparse Tuples to Deep Learning & Neuromorphic AI

- **Application Intelligence (Hearth):** Built-in algorithms and AI components turn raw asynchronous event streams into actionable semantic data.
- **Reduced Computational Load**: Because redundant background data is naturally filtered out at the sensor level, downstream processors handle significantly smaller data loads.
- **Advanced Tracking & Measurement**: Facilitates real-time detection, tracking, and monitoring of high-speed transient anomalies without heavy computational framing loops.

Timeline

Origins and Discovery of Event-Based Vision

Foundational Pioneers of the Silicon Retina

1988

Caltech Silicon Retina: Pioneered by Misha Mahowald and Carver Mead, who built the first three-layer Kuffler retina on a CMOS chip using logarithmic photoreceptors and the Address-Event Representation (AER) protocol.

2002

First Asynchronous Time-Based Image Sensor: Tobi Delbrück and Patrick Lichtsteiner at the Institute of Neuroinformatics (INI, University of Zurich and ETH Zurich) began developing the architecture that led to the modern Dynamic Vision Sensor (DVS).

2008

The DVS128 Breakthrough: Patrick Lichtsteiner, Christoph Posch, and Tobi Delbruck at ETH Zurich / UZH introduced self-timed switched-capacitor differencing, achieving 120 dB dynamic range and 15 µs latency.

Origins and Discovery of Event-Based Vision

Integration of Algorithms and SNNs

2011

Event-Based Algorithm Paradigm:
Researchers developed new mathematical tools tailored for event streams, including Time Surfaces, Event-Based Hough Transforms, and asynchronous clustering for tracking high-speed objects

2015

Integration with Spiking Neural Networks (SNNs):
As neuromorphic hardware (like IBM's TrueNorth and Intel's Loihi) emerged, researchers began bridging event-based cameras with SNNs, achieving ultra-low-power, bio-realistic pattern recognition and gesture classification.

2017

DAVIS and ATIS Commercialization:
Sensors combining standard active-pixel frames (APS) with asynchronous event-based streams (DAVIS - Dynamic and Active-pixel Vision Sensor) became widely available to researchers, bridging the gap between traditional and event-based vision.

Origins and Discovery of Event-Based Vision

Modern Day Implementation and Research

2021-23

Hybrid Transformers and Deep Learning for Events:
The field moved toward deep learning architectures adapted for event data.
For example: Event-based transformers and conversion of CNNs to SNNs

2024

Commercialization in Consumer and Automotive Sectors:
Event sensors started seeing integration into tech like smartphones, AR/VR headsets and ADAS for high emergency braking scenarios.

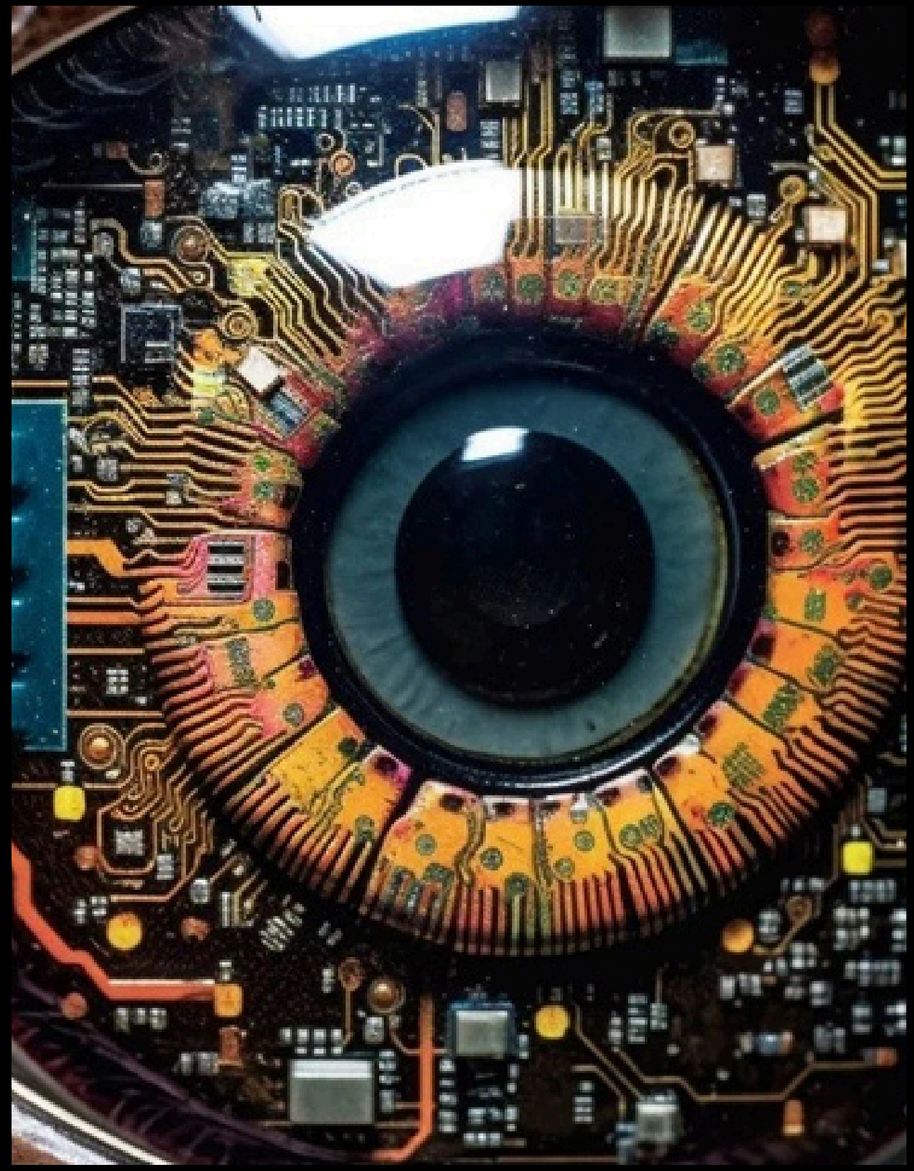
Present Day

Event-based vision is established as a critical pillar of edge AI and neuromorphic engineering. Current research focuses on unifying asynchronous event streams with standard RGB frames, scaling up Spiking Neural Network accelerators, and achieving sub-millisecond perception in extreme lighting conditions

Event-Camera Hardware and Architecture

Modern event sensors combine 3D-stacked analog pixel arrays with asynchronous digital arbitration and hardware noise filtering.

- **Independent Pixel Design**: Each pixel operates autonomously, containing its own local illumination monitoring and level-crossing detector circuits.
- **Patented Pixel Architecture**: Advanced silicon-level sensing design that optimizes high-speed photon collection alongside micro-watt power scaling.
- **Embedded AI & Processing Integration**: Integrates seamlessly with edge processing modules and software development kits for real-time interpretation.



Event Based Cameras v/s Frame Cameras

Key features	Event Based Camera	Frame Based Camera	Practical Advantage
Timing and Latency	Event cameras achieve 0.2 µs -1 µs to 150 µs internal clock resolution	A 60 fps camera gets 16.6 ms fram resolution while high speed cameras go till 0.1 ms	Sub-millisecond tracking of transient ballistic events without stroboscopic effects or frame gaps.
Dynamic Range & Motion Blur	Event sensors reach >120 dB to >140 dB without exposure motion blur.	~60 dB to 72 dB and severe shutter smear in standard frame cameras	Immune to direct solar washouts, tunnel exits, and headlamp glare.
Data Volume & Bandwidth	Scene-dependent sparse output achieves a 10x to 1,000x reduction in raw data volume.	Dense frame streaming.	Up to 99.9% data compression at source and scales bandwidth dynamically with scene activity.
Sensor Power Efficiency	Bare event sensor dies consume 36 µW (sleep) to 3–80 mW (active), peaking around 150mW.	~1.5 -4.5W for conventional camera modules. Going upto 40W in peak conditions.	10x to 100x longer battery life for micro-UAVs, smart glasses, and IoT edge nodes.
Visual Comparison	Continuous DVS event stream.	High-speed Photron 7.5 kHz frames	Feature corners and spatial edges stay sharp at velocities >1000°/s.

Real-World Applications

Robotics and Autonomous Navigation

High-Speed Obstacle Avoidance & Projectile Catching

Callout 1 — Quadrotor Dodgeball (Science Robotics 2020): Using onboard stereo event cameras and spatiotemporal clustering, Falanga et al. achieved 3.5 ms perception latency, dodging balls thrown at 10 m/s where 30–60 fps cameras (>30 ms latency) fail.

Callout 2 — Bio-Inspired UAV Flight (2025–2026): Ganglionic center-surround models cut UAV avoidance latency to 2.3 ms at 10 m/s (NeurIPS 2025), while the OmniRisk framework (2026) dodges obstacles at 15 m/s.

Callout 3 — Legged Robot Ball Catching (ICRA 2023): Forrai et al. deployed an event camera on an ANYmal quadrupedal robot to track and catch fast airborne projectiles despite walking vibration.

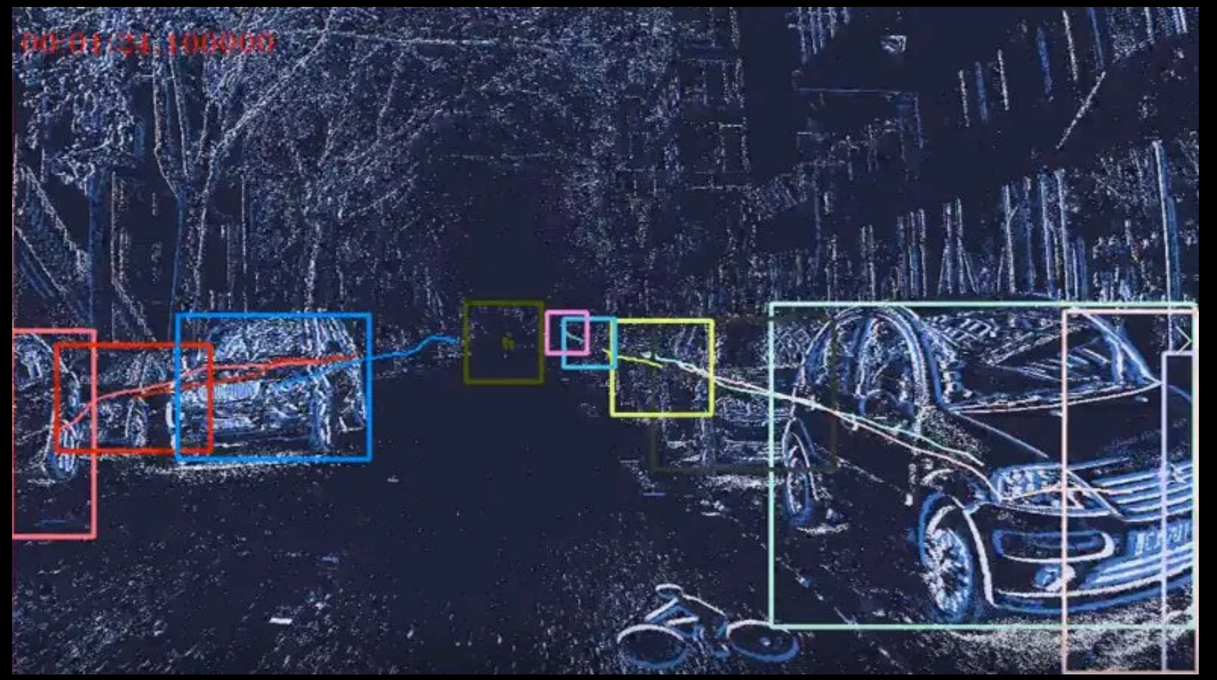
Automotive Technology and Driver Assistance

High-Speed Hazard Detection in Extreme Lighting

Tunnel Exits & Night Glare: Vehicles encountering blinding sunlight at tunnel exits or oncoming headlights at night face >100 dB illumination ratios that saturate or underexpose standard RGB cameras.

DSEC Stereo Driving Benchmark: Gehrig et al. (IEEE RA-L 2021) demonstrated on the DSEC dataset that event streams preserve pedestrian and road boundaries lost in standard frames.

5,000 Hz Latency at 45 Hz Bandwidth: Fusing a 20 Hz RGB camera with an event camera matches the obstacle-detection latency of a 5,000 Hz high-speed camera at the data bandwidth of a 45 Hz camera.



Industrial Automation & Quality Inspection

Non-Contact Vibration Monitoring & High-Speed Inspection

Full-Field Vibration Mapping: DVS systems measure machinery vibration frequencies from 1 Hz into the kHz range across every pixel simultaneously with 1-pixel spatial accuracy.

High-Speed Counting & Spatter Tracking: Microsecond timestamping enables blur-free monitoring of fast bottling lines, laser welding spatter, and sorting chutes without high-wattage strobe lights.

Event-Guided Motion Deblurring: Subtracting the double integral of the event stream from a blurred RGB frame (Pan et al., CVPR 2019) improves dynamic visual reconstruction by >50%.

Healthcare, Biomedical Imaging, and Edge Research

Experimental Instrumentation vs. Clinical Deployment

High-Speed Microfluidic Velocimetry: Tracks rapid particle flow in laboratory microfluidic channels at microsecond resolution without terabytes of high-speed video storage or intense lighting that damages biological samples.

GPU-Free Posture & Fall Monitoring (2026): Timestamp-recency encoding (T_s = 100ms) achieves 90.8% accuracy and an 11x speed-up for smart-home human posture recognition on edge CPUs

Privacy-Preserving Edge Sensing: Because raw event streams capture moving edge contours rather than recognizable static facial textures, they support privacy-conscious patient monitoring in experimental care settings.

Virtual Reality, Augmented Reality, & HCI

Extended Reality (XR) headsets demand kilohertz eye and hand tracking within strict milliwatt battery envelopes

- **1,000 Hz Wearable Eye Tracking at 20 mW:** Powered by the Prophesee GenX320, the 7invensun XR module tracks rapid saccadic eye movements at 1,000 Hz (1 ms updates) within a 20 mW total system power budget.

- **30x Lower Bandwidth 3D Motion Capture: Event-based 3D human motion capture at 1,000 Hz operates with 30x less data bandwidth than high-speed frame-based cameras.**

- **Sub-Millisecond Gesture Control: Pixel latency below 150 µs enables near-zero-lag finger and hand gesture interfaces for interactive AR glasses**

Aerospace, Security, and Space Situational Awareness

Space Situational Awareness & Planetary Landing

The Astrosite Neuromorphic Telescope: Western Sydney University's ICNS built Astrosite, a mobile neuromorphic observatory that tracks satellites and orbital debris across both daytime and nighttime skies.

International Space Station (ISS) Deployment: **Event sensors aboard** the ISS monitor orbital debris and high-speed atmospheric phenomena under extreme solar glare without saturating.

ESA Eventful Landings Project: The European Space Agency (ESA) utilizes event-based optical flow for autonomous spacecraft hazard avoidance during lunar and Martian descent.

Performance and Critical Evaluation

Quantitative Technical & Performance Comparison

Verified Specifications Across Sensor Generations vs. Frame CMOS

INI DVS128 (2008) & DAVIS240 (2014): DVS128 (128 x 128) achieves 15 µs latency, 3 kHz bright bandwidth, 120 dB DR, and ~24 mW power; hybrid DAVIS240 (240 x 180) reaches 3 µs latency and 130 dB DR.

Sony-Prophesee IMX636 (2020–2026 HD): 1280 x 720 resolution (4.86µm stacked pixel), <100 µs latency at 1,000lux, >120 dB DR (80mlux–100klux), and 80 mW active sensor power (2.2 mW standby).

Prophesee GenX320 (2025 Embedded): 320 x 320 resolution (6.3µm pixel), 1 µs timestamping, <150 µs latency at 1,000 lux, >140 dB DR (10mlux–100 klux), 36 µW sleep, and 3 mW active power.

Standard 60 fps Frame Camera Baseline: VGA to multi-megapixel (3.45 µm typical pitch), 16.7 ms frame interval (30 Hz Nyquist limit), ~60 dB DR, and ~1 W typical camera module power.

Illumination Scaling, Power States, & Competing Tech

How Illumination & System Packaging Govern Real-World

- **Low-Power Footprint:** Consumes minimal power (<10mW) due to event-driven activation.
- **Adaptability Across Lighting:** Adapts dynamically to extreme light-to-dark transitions without requiring manual exposure or gain adjustments.
- **Prophesee Product Ecosystem:** Built around core technologies like the Embra HD exploration kit, Mantra (defense/drone tracking), and Nexara (industrial automation).

Pros & Cons

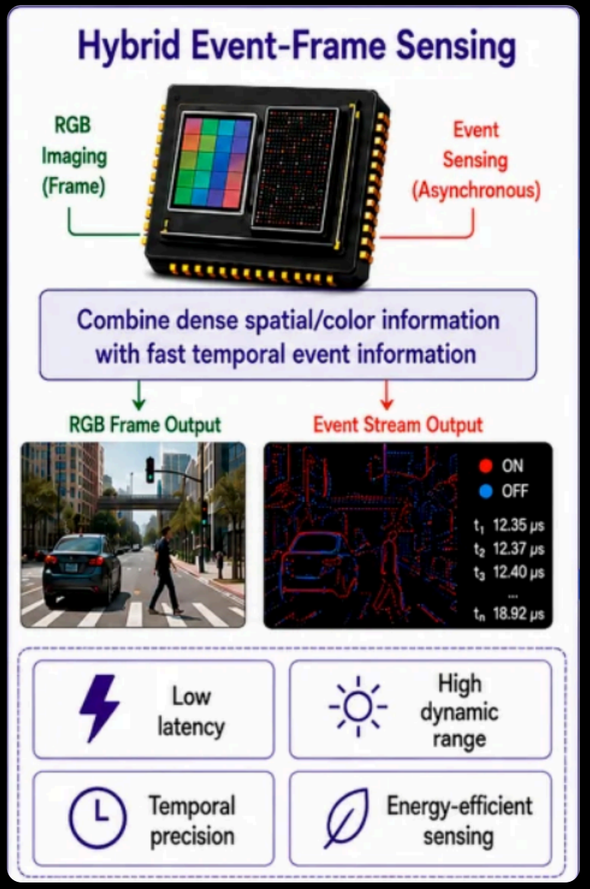
Key features	Pros	Cons
Temporal Resolution	Microsecond timestamp precision which captures dynamics >10,000 FPS equivalent.	System processing bottlenecks can negate sensor-level speed.
Motion Blur	Edges of fast-moving objects remain sharp instead of blurring out.	Extreme low light can cause photodiode lag, reintroducing temporal smearing.
Data Efficiency	>90% data reduction in sparse, static scenes	"Event storms" in highly textured, moving scenes can exceed standard video bandwidth.
Dynamic Range	>120-140 dB; sees into shadows and bright sun simultaneously.	Contrast sensitivity drops near absolute darkness.
Power Consumption	Extremely low power (~3-80 mW) in static/sparse environments.	Power draw scales linearly with scene activity (event rate).
Spatial / Static Vision	N/A	Blind to static scenes or low spatial resolution (~1 MP).
Software Ecosystem	Enables ultra-fast Spiking Neural Networks (SNNs).	Incompatible with standard Convolutional Neural Networks (CNNs) without formatting overhead.

Future of Event-Based Vision

Hybrid Multimodal Sensors & Neuromorphic Edge Processors (2025–2026)

- **Hybrid In-Pixel & Multi-Sensor Fusion:** Combining global-shutter frames and events (from DAVIS240 to 2026 DAPEVO and EventHub frameworks) merges static RGB texture with microsecond motion.

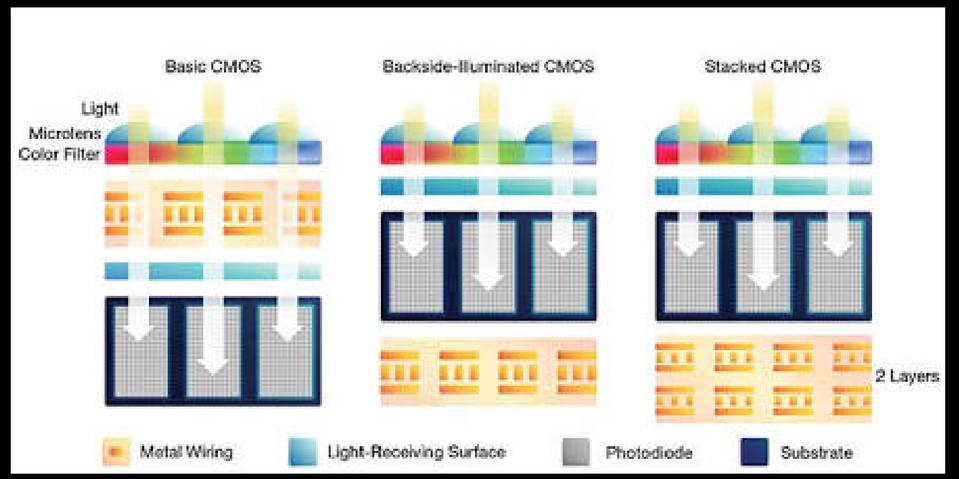
- **Sub-30 µs Neuromorphic Processors:** The 2026 ETHEREAL Event-Driven Graph Neural Network chip achieves 25.6 µs inference latency at 1.6–1.7 µJ/event on VGA workloads—a 10x to 1,000x leap.
- **Embodied Humanoid & Space Perception:** Fusing event sensors with SPAD single-photon sensors and foundation models unlocks low-power vision for humanoid teleoperation and deep-space landers.

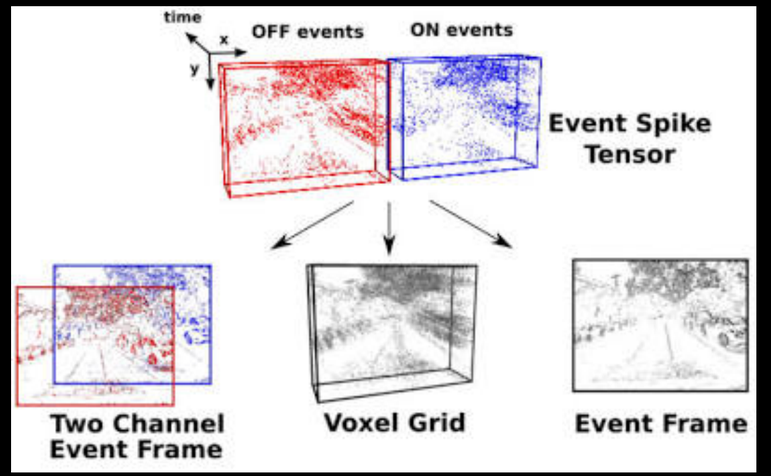


Barriers to Mainstream Adoption

Engineering & Ecosystem Challenges to Overcome 3D-Stacked Wafer

- **3D-Stacked Wafer Manufacturing Cost:** Bonding back-illuminated photodiodes to per-pixel CMOS analog comparators increases fabrication complexity compared to commodity smartphone sensors.
- **Software & Deep Learning Ecosystem Gap:** Mainstream computer vision libraries and GPU pipelines are optimized for dense synchronous grids rather than sparse (x,y,t,p) tuples.
- **High-Event-Rate Bus Saturation:** Under extreme full-frame texture motion or unmitigated lighting flicker, event rates can saturate readout buses (requiring MIPI CSI-2 compression and hardware AFK/STC tuning).
- **Benchmark Standardization:** Because performance varies with lux, bias tuning, and motion speed, standardized datasets (DSEC, ECCV 2026 SEE Challenge) are essential for fair evaluation.





Conclusion

- By abstracting the mammalian retina's logarithmic photoreceptor adaptation and transient ON/OFF ganglion pathways into CMOS silicon, event cameras replace synchronous frames with continuous-time change detection.

- Event cameras are complementary partners to conventional RGB cameras—transformative whenever applications demand microsecond reaction times (3–150 µs), extreme dynamic range (>120–140 dB), motion-blur immunity, and milliwatt edge efficiency (36 µW–80 mW).

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Thank You!

Any Questions?